Section 2

Systems Programming ECE3502

SIC machine architecture

Memory

it consists of 8-bit bytes

3 consecutive byte from a word(24 bits)

Total of 32768 (2^15) bytes in the computer memory

Data Formats

Integers are stored as 24-bit binary numbers

2'S complement representation is used for negative values

Characters are stored using their 8-bit ASCII codes

No floating-point hardware on the standard version of sic

SIC machine architecture

Instruction formats

24-bit format

opcode (8)	x	address (15)

X is to indicate index-address mode

Addressing modes

Mode	Indication	Target address calculation	
Direct	x=0	TA = address	**LDA ALPHA → MEMORY[ALPHA]**
Indexed	x=1	TA = address + (X)	**LDA ALPHA,X → MEMORY[ALPHA+X]**

Assembler Directives

Pseudo-Instructions

- Not translated into machine instructions
- Providing information to the assembler

Basic assembler directives

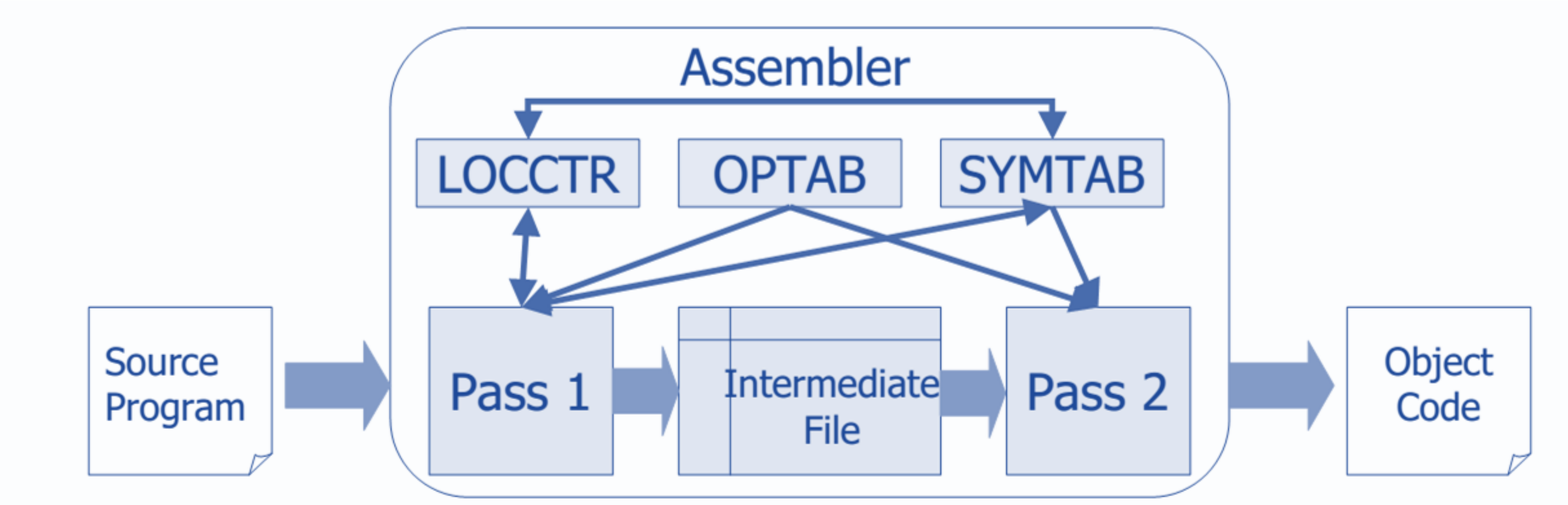
- **START:** Specify name and starting address for the program
- **END:** Indicate the end of the source program, and (optionally) the first executable instruction in the program.
- **BYTE:** Generate character or hexadecimal constant, occupying as many bytes as needed to represent the constant.
- **WORD:** Generate one-word integer constant
- **RESB:** Reserve the indicated number of bytes for a data area
- **RESW:** Reserve the indicated number of words for a data area

Assembler Data Structures

Operation Code Table (OPTAB)

Symbol Table (SYMTAB)

Location Counter (LOCCTR)



Pass One

PASS ONE:

1.Location Counter

Instruction FORMAT	+3
RESW	3*# OF RESERVED WORDS
RESB	1*# OF RESERVED WORDS
WORD	+3
BYTE	X' ' each 2 hex = 1 byte C' ' count and add

2.Symbol Table (each symbol with its location)

Pass One

Sheet 2 Question 1

Translate (by hand) the following assembly program to SIC object code. Which contains H record, T record, and E record.

Location Counter	Symbols	Instructions	Reference
0000	Prog1	Start	0000
0000		LDA	ZERO
0000+3=0003		STA	INDEX
0003+3=0006	LOOP	LDX	INDEX
0006+3=0009		LDA	ZERO
0009+3=000C		STA	ALPHA, X
000C+3=000F		LDA	INDEX
000F+3=0012		ADD	THREE
0012+3=0015		STA	INDEX
0015+3=0018		COMP	K300
0018+3=001B		TIX	TWENTY
001B+3=001E		JLT	LOOP
001E+3=0021	INDEX	RESW	1
0021+3=0024	ALPHA	RESW	100
$0024+(100*3)^{10} = 0150$	ZERO	WORD	0
0150+3=0153	K300	WORD	100
0153+3=0156	THREE	WORD	3
0156+3=0159	TWENTY	WORD	20
0159+3=015C	End	0000	

Symbol Table

Symbol	Address
Prog1	0000
Loop	0006
Index	0021
Alpha	0024
Zero	0150
K300	0153
Three	0156
Twenty	0159

Pass Two

Question 1

Location Counter	Symbols	Instructions	Reference	Obj. Code
0000	Prog1	Start	0000	No obj. code
0000		LDA	ZERO	000150
0003		STA	INDEX	0C0021
0006	LOOP	LDX	INDEX	
0009		LDA	ZERO	
000C		STA	ALPHA, X	
000F		LDA	INDEX	
0012		ADD	THREE	
0015		STA	INDEX	
0018		COMP	K300	
001B		TIX	TWENTY	
001E		JLT	LOOP	
0021	INDEX	RESW	1	
0024	ALPHA	RESW	100	
0150	ZERO	WORD	0	
0153	K300	WORD	100	
0156	THREE	WORD	3	
0159	TWENTY	WORD	20	
015C	End	0000		

LDA ZERO		
Opcode (8)	X	Address (15bits)
00	0	0150
0000 0000	0	~~0~~000 0001 0101 0000
00	0150	

STA INDEX		
Opcode (8)	X	Address (15bits)
0C	0	0021
0000 1100	0	~~0~~000 0000 0010 0001
0C	0021	

Pass Two.

Question 1

Location Counter	Symbols	Instructions	Reference	Obj. Code
0000	Prog1	Start	0000	No obj. code
0000		LDA	ZERO	000150
0003		STA	INDEX	0C0021
0006	LOOP	LDX	INDEX	040021
0009		LDA	ZERO	000150
000C		STA	ALPHA, X	0C8024
000F		LDA	INDEX	00 0021
0012		ADD	THREE	18 0156
0015		STA	INDEX	0C 0021
0018		COMP	K300	28 0153
001B		TIX	TWENTY	2C 0159
001E		JLT	LOOP	38 0006
0021	INDEX	RESW	1	No obj. code
0024	ALPHA	RESW	100	No obj. code
0150	ZERO	WORD	0	000000
0153	K300	WORD	100	000064
0156	THREE	WORD	3	000003
0159	TWENTY	WORD	20	000014
015C	End	0000		No obj. code

LDX INDEX		
Opcode (8)	X	Address (15bits)
04	0	0021
0000 0100	0	~~0~~000 0000 0010 0001
04	0021	

STA ALPHA, X		
Opcode (8)	X	Address (15bits)
OC	1	0024
0000 1100	1	~~0~~001 0000 0010 0100
0C	8024	

Pass Two..

Question 1

H.T.E Record

H. program name(6). Starting address(6). Length(6)

H.XXProg1.000000.00015C

T. Starting address of each t record(6) . Size of instruction(2). obj. code of all inst(6)

T.000000.1E.000150.0C0021.040021.000150.0C8024.00021.180156.0C0021.280153.2C0159

T.00001E.03.380006

T.000150.0C. 0000000.000064.0000003.000014

E. Starting address(6)

E.000000

More Examples

Instructions	Reference	Obj. Code
BYTE	X ’05’	05
BYTE	C ’EOF’	454F46

PASS ONE & TWO

Question 2

Location Counter	Symbols	Instructions	Reference	Obj. Code
0030	HW1B	START	0030	No Obj. code
0030	WRREC	LDX	ZERO	04 0045
0033	WLOOP	TD	OUTPUT	E0 004B
0036		JEQ	WLOOP	30 0033
0039		LDCH	RECORD, X	50 804C
003C		WD	OUTPUT	DC 004B
003F		TIX	LENGTH	2C 0048
0042		JLT	WLOOP	38 0033
0045	ZERO	WORD	0	000000
0048	LENGTH	WORD	1	000001
004B	OUTPUT	BYTE	X ‘05’	05
004C	RECORD	RESB	100	No Obj. code
00B0	End	0030		No Obj. code

Symbol	Address
WRREC	0030
WLOOP	0033
ZERO	0045
LENGTH	0048
OUTPUT	004B
RECORD	004C

H.HW1BXX.000030.000080

T.000030.1C.040045.E0004B.300033.

50804C.DC004B.2C0048.380033.000000.

000001.05

E.000030